GLOBAL ACADEMY OF TECHNOLOGY

Department of Information Science & Engineering

Mini Project Presentation

Online Quiz Web App

Guided By: Dr. Vimuktha ES

Presented By: **HARISH GOWDA B**
YASHAS KY

Agenda

1. Overview / Background
2. Motivation
3. Objectives
4. Literature Review
5. Problem Statement
6. Project Requirements
7. System Architecture
8. Implementation
9. Experimental Results
10. Conclusion
11. References

Overview / Background

Introduction

An Online Quiz Application is a web-based platform designed to create, manage, and take quizzes.

Purpose

It serves educational institutions, corporate training programs, and general users for testing knowledge.

Users

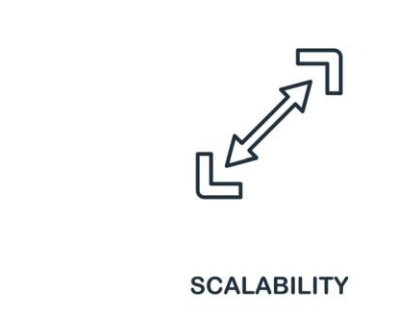
Targets students, educators, trainers, and quiz enthusiasts

Technology

Utilizes web technologies like HTML, CSS, JavaScript for the frontend and Node.js, Django, or Flask for the backend.

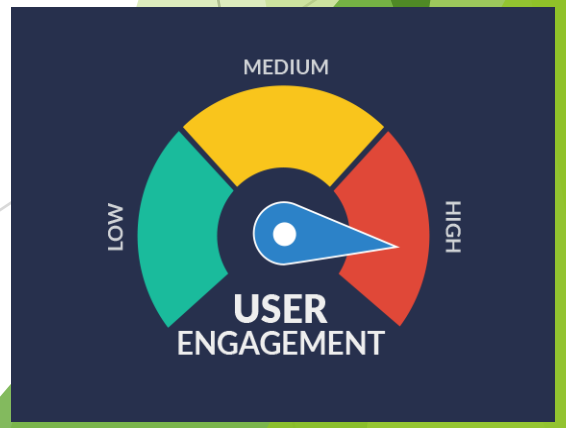
Motivation

Traditional Methods: Paper-based quizzes are labor-intensive, requiring manual grading and are slow to provide feedback to students.



Limited Accessibility : Physical quizzes are confined to specific locations and times, limiting convenience for users.

Scalability Issues: Handling and administering quizzes for a large number of participants manually is inefficient and prone to errors.



Engagement: Traditional quizzes often lack interactive elements, which can lead to lower engagement and motivation among users.

Objectives

User-Friendly Interface

Develop a simple and intuitive interface that is easy for anyone to use.

Instant Feedback

Provide automatic scoring and immediate feedback to help users learn and improve quickly.

Secure Access

Implement strong security measures to protect user data and ensure safe access

Device Compatibility

Ensure the application works seamlessly on various devices, including computers, tablets, and smartphones.

Literature Review

Literature Review

Author	Title	Journal	Year	Approach	Key Findings
1.Sadhvi D Alva 2.Mohammed Aiman 3.Mohammed Zaid 4.Ajmal Abdul Rahiman	ONLINE QUIZ EXAM PORTAL	International Journal of Innovative Science and Research Technology	2023	The Online Quiz Exam Portal replaces traditional paper-based exams with a digital platform. Students can take quizzes online, eliminating the need for paper and pencil. The system uses a rule-based algorithm to grade quizzes instantly and accurately. This saves time for both students and educators while ensuring secure and precise results.	Efficiency: Automates grading to save educators time. Accessibility: Available anytime, anywhere. Immediate Feedback: Provides instant results. Security: Ensures exam integrity with centralized control.
1.M Divya Sri 2.Poojithanjali 3.G. Nikhil Harsha 4.M. Reshwanth	Online Quiz Website	International Journal of Advanced Research in Science, Communication and Technology (IJARSCT)	2022	Technology Stack: Uses Visual Studio Code, HTML, CSS, PHP, XAMPP, and JavaScript. Database: MySQL integrated with PHP for storing quiz data and user info. User Interface: Simple and intuitive design for easy navigation and use. Admin Control: Admin panel for adding and managing quizzes. Security: Secure data handling and user authentication to prevent cheating.	Efficiency: Automates grading for time savings and instant results. Competitive Learning: Provides immediate feedback to motivate students. Accessibility: Available anytime, anywhere for convenience. Security: Ensures data protection and prevents cheating.
1.Syazwanie Filzah binti Zulkifli 2.Cheah Waishiang 3.Muhammad Asyraf bin Khairuddin 4.Nurfauza binti Jali 5.Yanti Rosmunie binti Bujang	How to Model an Engaging Online Quiz? The Emotion Modeling Approach	Journal of telecommunications and information technology	2022	Emotion Modeling: Integrates emotional goals to enhance engagement. Agent-Oriented Modeling (AOM): Models emotions in design, showing agent interactions. Emotion-Oriented Goals: Addresses both positive and negative emotions in development. Tropos Model: Analyzes how role interactions influence emotions.	Feasibility: AOM works well for modeling emotional needs and prototyping. Engagement: Emotional goals boost user engagement. Satisfaction: Quiz Master outperforms traditional quizzes in user satisfaction. Future: More research needed to refine the approach.

Problem Statement

Problem Statement

- Paper-based quizzes are slow, hard to access, and not engaging.
- They cause delays in feedback, limit participation, and have higher risks of errors and cheating.
- We need an online quiz application to solve these problems.

What is the difference between a Google quiz and a dedicated Online quiz portal?

Google Quiz

1. **Ease of Use:** Simple to create and share; ideal for quick, basic quizzes.

2. **Platform Integration:** Seamlessly integrates with other Google Workspace tools.

3. **Cost:** Generally free to use with a Google account.

4. **Customization:** Limited options for design and functionality customization.

5. **Data Handling:** Basic data handling and analytics features

Dedicated Online Quiz Portal

1.Advanced Features: Offers timed quizzes, random question generation, and multimedia support.

2.Customization: High level of customization for design, user interface, and functionality.

3.Data Analytics: Provides advanced data analytics and reporting capabilities.

4.Access Control: Robust user management and authentication options.

5.Security: Enhanced security features tailored for educational and corporate environments.

Project Requirement

Hardware / Software Requirements Identified

Hardware Requirements

Server

Type: Cloud-based (e.g., AWS, Heroku) or physical server
Specifications: Adequate CPU, RAM, and storage based on user load
Backup: Reliable backup and recovery solutions

Client Devices:

Types: Desktops, laptops, tablets, smartphones
Requirements: Must run modern web browsers and support internet connectivity
Screen Compatibility: Should work well on various screen sizes and resolutions

Software Requirements

Development Tools: Visual Studio Code or similar IDE for coding and project management.

Frontend Technologies: HTML, CSS, and JavaScript for designing and building the user interface.

Backend Technologies: PHP or another server-side language for handling server logic and interactions.

Web Server: XAMPP or similar software for running the web server locally during development.

Database: MySQL for storing and managing application data, including quizzes and user information.

Security Tools: SSL/TLS certificates to ensure secure data transmission and protect user information.

System Architecture

System Architecture

Client-Side: Uses HTML, CSS, and JavaScript to create the user interface and handle interactions.

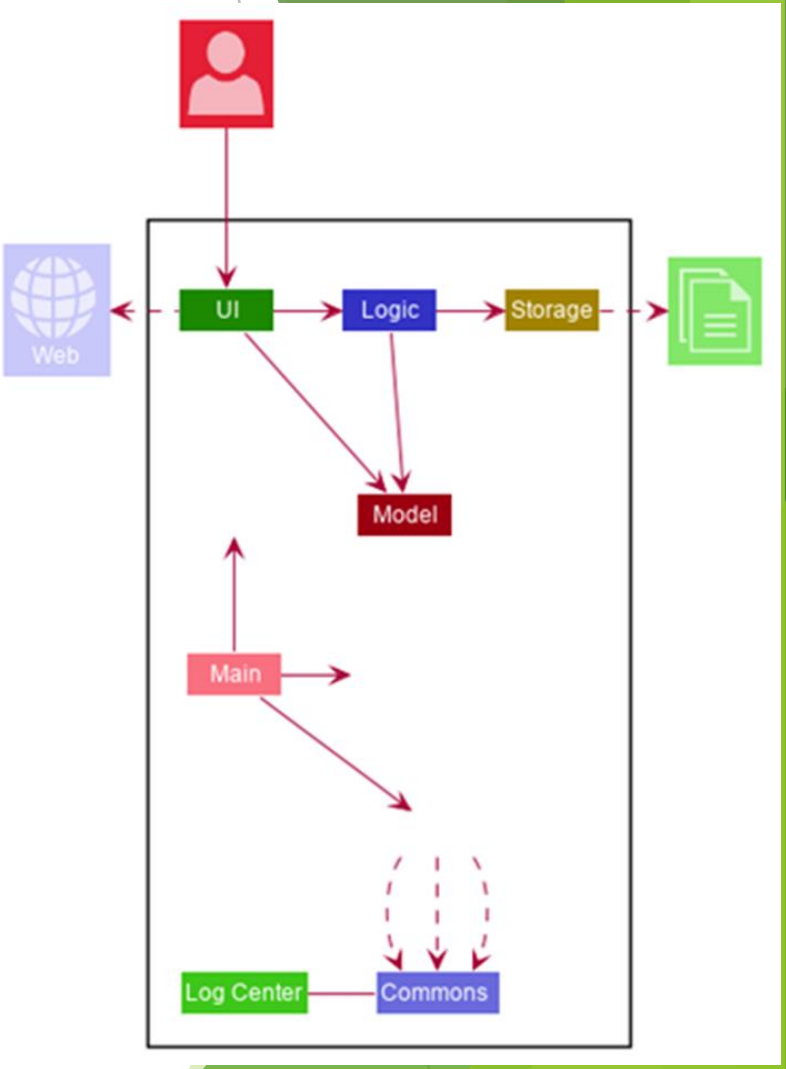
Server-Side: Runs on a web server (e.g., XAMPP) with server-side scripting (e.g., PHP) to process requests and manage data.

Database: MySQL stores and manages quiz and user data.

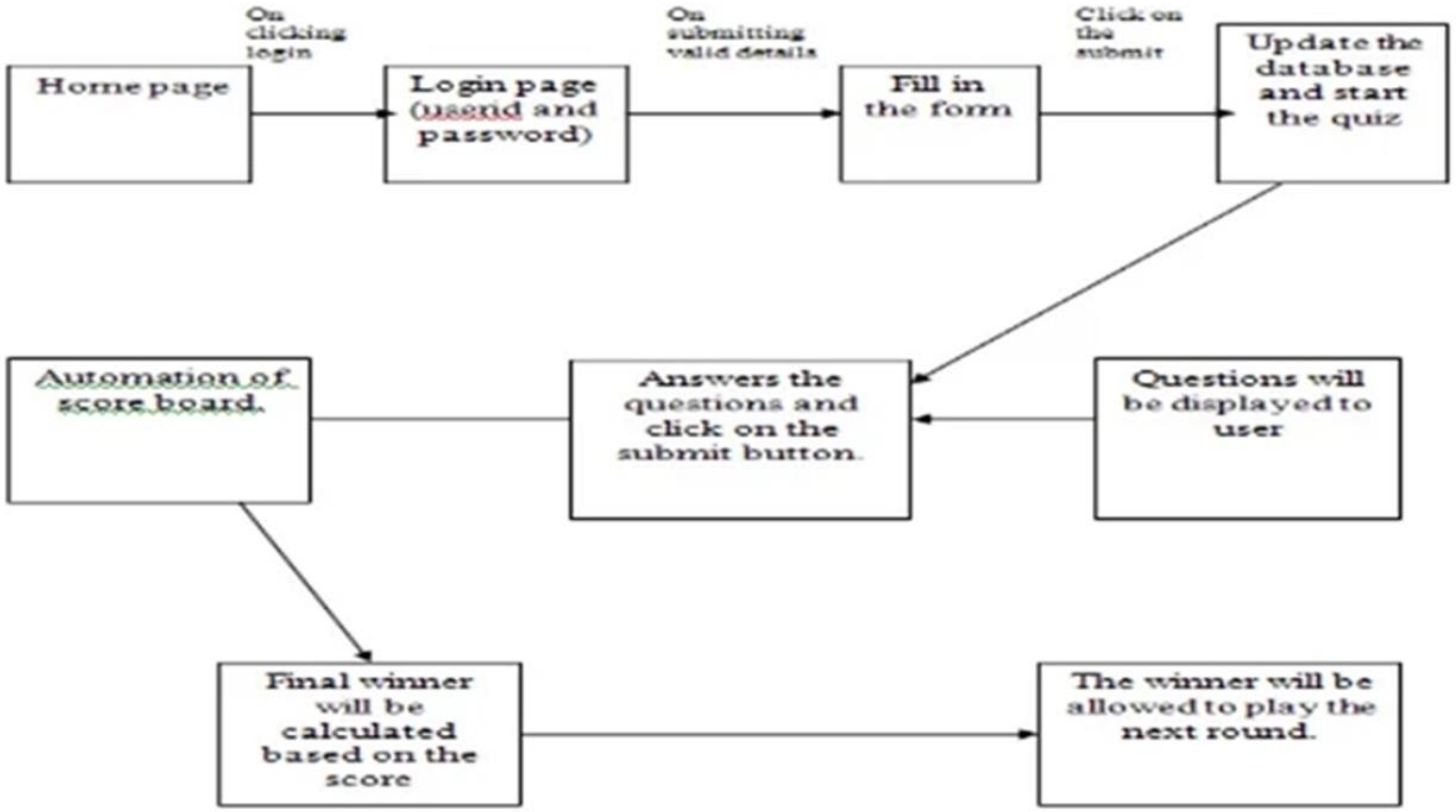
Communication: Uses HTTPS for secure data transfer; JSON or XML for data exchange.

Security: SSL/TLS certificates ensure secure data transmission.

Hosting: Deployed on cloud services (e.g., AWS) or physical servers



Proposed System Architecture



Implementation

1.Frontend Development:

- **Technologies**: HTML, CSS, JavaScript
- **Description**: Designed using HTML for structure, CSS for styling, and JavaScript for interactivity.

2.Backend Development:

- **Technologies**: Python, Flask
- **Description**: Flask manages routing, processes inputs.

3.Quiz Management Integration:

- **Technologies**: Speech Recognition API
- **Description**: Manages quiz creation, administration, grading, and user data storage.

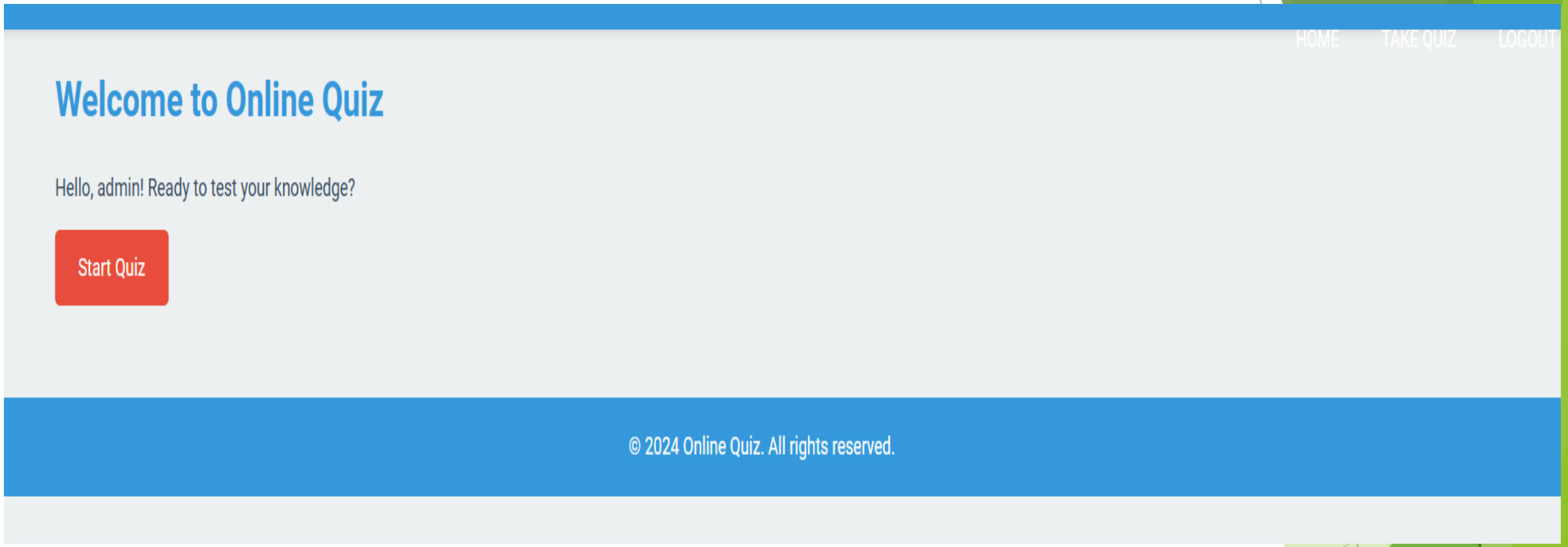
4.Response Handling:

- **Technologies**: JSON, Python Dictionary
- **Description**: User queries are matched against a Python dictionary, with responses formatted in JSON.

5.Interactive Features:

- **Technologies**: JavaScript, Flask
- **Description**: Instant feedback and dynamic question navigation using JavaScript on the frontend and Flask on the backend.

Experimental Results


HOME
TAKE QUIZ
LOGOUT
Welcome to Online Quiz
Hello, admin! Ready to test your knowledge?
Start Quiz
© 2024 Online Quiz. All rights reserved.

Register

Username:

Password:

Confirm Password:

Register

Already have an account? Login here

© 2024 Online Quiz. All rights reserved.

Login

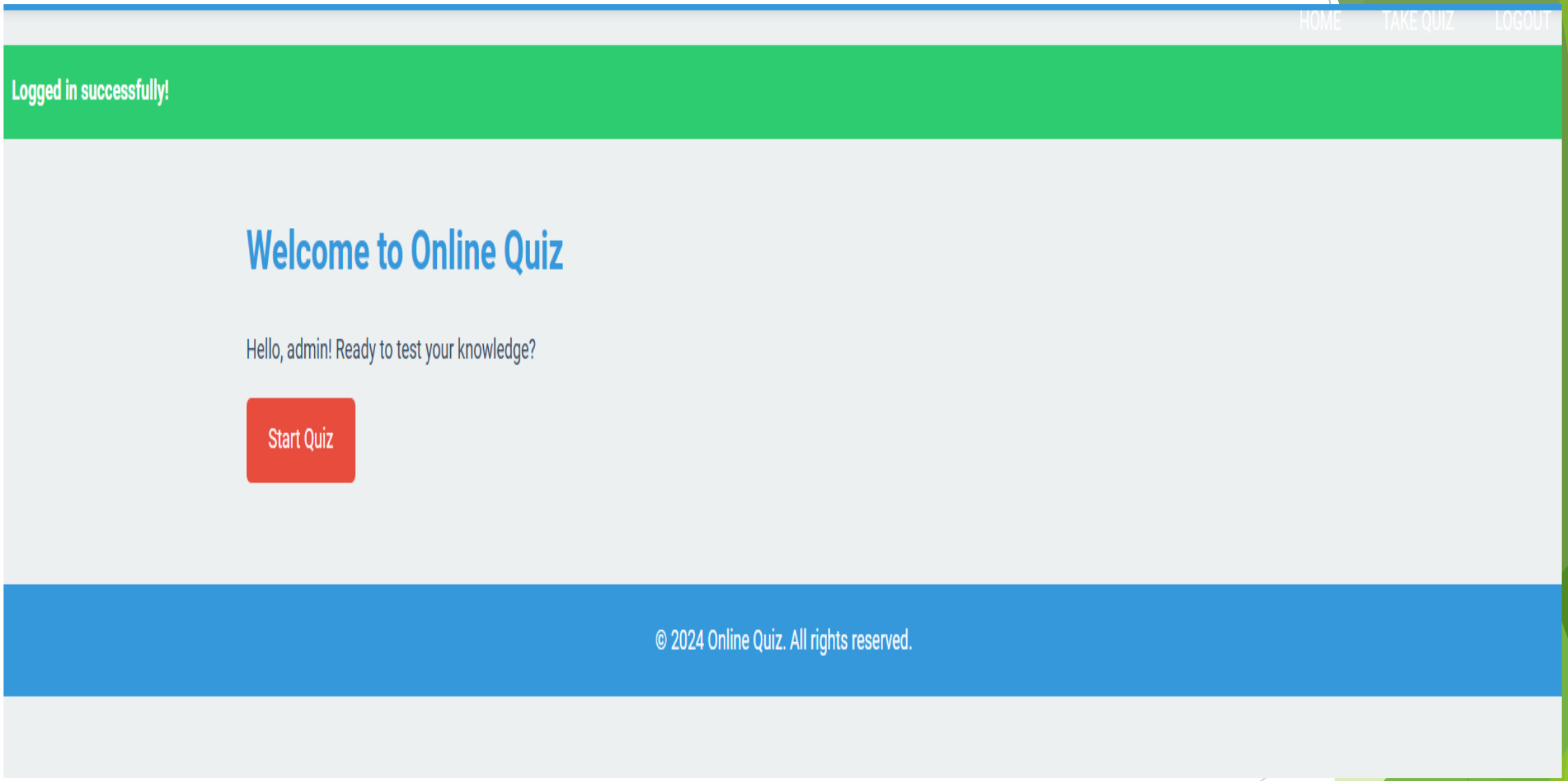
Username:

Password:

Login

Don't have an account? Register here

© 2024 Online Quiz. All rights reserved.


HOME
TAKE QUIZ
LOGOUT
Logged in successfully!
Welcome to Online Quiz
Hello, admin! Ready to test your knowledge?
Start Quiz
© 2024 Online Quiz. All rights reserved.

Quiz

Time left: 4:55

What is the capital of France?

- London
- Berlin
- Paris
- Madrid

Which planet is known as the Red Planet?

- Venus
- Mars
- Jupiter
- Saturn

Quiz

Time left: 4:09

What is the capital of France?

- London
- Paris
- Madrid

Please select one of these options.

What is the chemical symbol for gold?

- Ag
- Fe
- Au
- Cu

Submit Quiz

Quiz Results

Your score: 3 out of 5

Detailed Results

Take Another Quiz | Back to Home

© 2024 Online Quiz. All rights reserved.

Conclusion

•**Technologies Used:**
•Backend: Python, Flask for efficient processing.
•Frontend: HTML, CSS, JavaScript for a responsive interface.

•**Quiz Functionality**:
•Users answer multiple-choice questions through a simple web form.
•Flask processes submissions and provides instant feedback.

•**Data Handling**:
•JSON and Python dictionaries manage questions and answers, simplifying updates without a database.

•**User Experience**:
•Lightweight, user-friendly, and engaging quiz platform.
•Smooth interaction without complex AI or heavy database integrations.

References

"**Online Quiz Website**"
M Divya Sri, G Poojithanjali, G. Nikhil Harsha, M. Reshwanth.
International Journal of Advanced Research in Science, Communication and Technology,2022.

"**How to Model an Engaging Online Quiz? The Emotion Modeling Approach**"
Syazwanie Filzah binti Zulkifli, Cheah Wai Shiang, Muhammad Asyraf bin Khairuddin,Nurfauza Jali, Yanti Rosmunie binti Bujang.
Journal of Telecommunications and Information Technology,2022.

"**Online Quiz Exam Portal** "
Sadhvi D Alva, Mohammed Aiman, Mohammed Zaid, Ajmal Abdul Rahiman, S Divyashree.
International Journal of Innovative Science and Research Technology,2023.

THANK YOU